

Plant Physiology and Metabolism

Plant physiology and metabolism explain how plants absorb water and minerals, make food, transport materials, respire, grow, respond to stimuli, and maintain life. For BSc Botany, this is one of the most important advanced chapters because it connects structure with function and prepares you for ecology, biotechnology, agriculture, and research.

1. Introduction

Plant physiology is the branch of botany that studies the functions of living plants.

Metabolism is the sum total of all chemical reactions occurring inside plant cells, including anabolic reactions that build compounds and catabolic reactions that break them down.

This chapter is essential because plants are not passive organisms; they regulate water balance, open and close stomata, synthesize complex organic molecules, and control growth through hormones and enzymes. A strong understanding of physiology is also needed for crop improvement and environmental science.

2. Meaning and scope

Plant physiology deals with:

- Absorption of water and minerals.
- Movement of substances inside the plant.
- Photosynthesis and respiration.
- Transpiration and stomatal regulation.
- Growth and development.
- Seed germination and dormancy.
- Plant hormones and stress responses.
- Metabolic pathways and energy transformations.

Metabolism includes:

- Primary metabolism, which is essential for survival.
- Secondary metabolism, which produces alkaloids, pigments, terpenoids, phenolics, tannins, and many useful compounds.

3. Water in plant life

Water is the most important constituent of plant cells and the medium in which most biochemical reactions occur. It maintains turgidity, supports cell expansion, enables transport, and participates directly in photosynthesis.

Properties of water

- Polar molecule.
- High specific heat.
- High latent heat of vaporization.
- Cohesion and adhesion.
- Excellent solvent.

These properties make water suitable for transport and thermal stability in plants.

4. Absorption of water

Water is absorbed mainly by root hairs from the soil. It enters the plant by osmosis because the cell sap of root hairs is usually more concentrated than the surrounding soil solution.

Pathways of water movement

- Apoplast pathway: through cell walls and intercellular spaces.
- Symplast pathway: through cytoplasm connected by plasmodesmata.
- Transmembrane pathway: across cell membranes repeatedly.

Factors affecting absorption

- Soil water availability.
- Concentration gradient.
- Temperature.
- Aeration.
- Root surface area.
- Transpiration pulls.

5. Ascent of sap

Ascent of sap is the upward movement of water and minerals from roots to leaves through xylem. The most widely accepted mechanism is the cohesion-tension theory.

Cohesion-tension theory

- Water molecules cohere to each other.
- Water molecules adhere to xylem walls.
- Transpiration from leaves creates negative pressure.
- This pulls water upward as a continuous column.

BrainStack

This theory explains the movement of water in tall trees without the need for a mechanical pump.

6. Mineral nutrition

Plants require essential mineral elements for growth, metabolism, and reproduction. These are obtained from the soil in ionic form.

Essential elements

- Macronutrients: N, P, K, Ca, Mg, S.
- Micronutrients: Fe, Mn, Zn, Cu, B, Mo, Cl, Ni.

Functions

- Nitrogen: proteins, nucleic acids, chlorophyll.
- Phosphorus: ATP, nucleic acids, phospholipids.
- Potassium: enzyme activation, stomatal movement.
- Magnesium: chlorophyll and enzyme activity.
- Iron: electron transport and chlorophyll synthesis.

Deficiency symptoms

- Chlorosis.
- Necrosis.
- Stunted growth.
- Poor flowering and fruiting.
- Abnormal leaf shape.

7. Nutrient uptake and transport

Mineral ions are absorbed by roots through active and passive processes. Active transport requires energy and is crucial when ions move against concentration gradients.

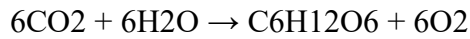
Transport of minerals

- Xylem transports water and minerals upward.
- Phloem transports food materials, especially sucrose, amino acids, and signaling molecules.

8. Photosynthesis

Photosynthesis is the process by which green plants use light energy to synthesize organic food from carbon dioxide and water, releasing oxygen.

Overall equation



Site

Photosynthesis occurs in chloroplasts, especially in mesophyll cells of leaves.

Pigments

- Chlorophyll a.
- Chlorophyll b.
- Carotenoids.
- Xanthophylls.

Phases

- Light reaction.
- Dark reaction or Calvin cycle.

9. Light reaction

The light reaction occurs in the thylakoid membranes of chloroplasts. It involves absorption of light energy, splitting of water, release of oxygen, and formation of ATP and NADPH.

Key events

- Photolysis of water.
- Electron transport.
- Photophosphorylation.
- Reduction of NADP to NADPH.

Types of photophosphorylation

- Non-cyclic.
- Cyclic.

10. Dark reaction

The dark reaction occurs in the stroma of chloroplasts and does not directly require light, though it depends on ATP and NADPH from the light reaction.

Calvin cycle steps

- Carboxylation.
- Reduction.
- Regeneration.

The enzyme RuBisCO plays a central role in carbon fixation.

11. C3, C4, and CAM pathways

Plants have different pathways of carbon fixation that reflect adaptation to environment.

C3 plants

- First stable product is 3-carbon compound.
- Most common pathway.
- Example: rice, wheat.

C4 plants

- First stable product is 4-carbon compound.
- Efficient in hot, bright conditions.
- Example: maize, sugarcane.

CAM plants

- Stomata open at night.
- Adapted to dry conditions.
- Example: cactus, pineapple.

12. Respiration

Respiration is the process by which plants break down organic molecules to release energy in the form of ATP.

Types

- Aerobic respiration.
- Anaerobic respiration.

Stages of aerobic respiration

- Glycolysis.
- Krebs cycle.
- Electron transport chain.

Respiration occurs in all living cells and is essential for growth, transport, biosynthesis, and active absorption.

13. Photorespiration

Photorespiration is a wasteful process in which RuBisCO acts as oxygenase instead of carboxylase, reducing photosynthetic efficiency.

Importance

- More prominent in C3 plants.
- Increases under high temperature and low CO₂.
- Less in C4 and CAM plants.

14. Transpiration

Transpiration is the loss of water vapor from aerial parts of the plant, mainly through stomata.

Types

- Stomatal transpiration.
- Cuticular transpiration.
- Lenticular transpiration.

Significance

- Cooling effect.
- Helps ascent of sap.
- Maintains turgor.
- Supports mineral transport.

Control

Stomatal opening and closing are regulated by guard cells, turgor pressure, light, CO₂ concentration, and abscisic acid.

15. Stomatal mechanism

Stomata are pores surrounded by guard cells. When guard cells absorb water and become turgid, stomata open. When they lose water and become flaccid, stomata close.

This mechanism is central to balancing CO₂ entry for photosynthesis and water loss by transpiration.

16. Plant growth

Growth is a permanent and irreversible increase in size, mass, and volume. It involves cell division, cell elongation, and cell differentiation.

Phases of growth

- Meristematic phase.
- Elongation phase.
- Maturation phase.

Growth curve

The sigmoid or S-shaped growth curve is common in plants.

17. Plant hormones

Plant hormones are chemical messengers that regulate growth and development.

Major hormones

- Auxins.
- Gibberellins.
- Cytokinins.
- Absciscic acid.
- Ethylene.

Auxins

Promote cell elongation, apical dominance, root initiation, and tropic responses.

Gibberellins

Promote stem elongation, seed germination, and flowering in some plants.

Cytokinins

Promote cell division and delay senescence.

Absciscic acid

Acts as a growth inhibitor, promotes dormancy, and closes stomata during stress.

Ethylene

Promotes fruit ripening, senescence, and abscission.

18. Tropic and nastic movements

Plants show movement responses to environmental stimuli.

Tropic movements

Directional growth responses.

- Phototropism.
- Geotropism.
- Hydrotropism.
- Thigmotropism.

Nastic movements

Non-directional responses.

- Seismonasty.
- Nyctinasty.

19. Seed germination

Germination is the resumption of growth in the embryo of a seed under favorable conditions.

Conditions required

- Water.
- Oxygen.
- Suitable temperature.
- Sometimes light.

Types

- Epigeal germination.
- Hypogeal germination.

20. Dormancy

Dormancy is a temporary resting state in seeds or buds in which growth is suspended even under favorable conditions. It prevents germination at unsuitable times.

21. Enzymes and metabolism

Enzymes are biological catalysts that speed up biochemical reactions. Plant metabolism depends on highly regulated enzyme systems.

Enzyme properties

- Specificity.
- High catalytic power.
- Sensitivity to pH and temperature.
- Often require cofactors or coenzymes.

Metabolic regulation

- Feedback inhibition.
- Allosteric control.
- Gene-level control.
- Compartmentation in organelles.

22. Carbohydrate metabolism

Carbohydrates are central to plant metabolism.

Key processes

- Photosynthetic synthesis of glucose.
- Starch formation and storage.
- Glycolysis.
- Krebs cycle.
- Pentose phosphate pathway.

Plants store carbohydrates mainly as starch, sucrose, and fructans depending on species.

23. Lipid metabolism

Lipids serve as structural and storage molecules.

Functions

- Membrane formation.
- Energy storage.
- Protective coating.
- Signaling molecules.

Fatty acids are synthesized in plastids and later used for membrane and reserve lipid formation.

24. Protein metabolism

Proteins are essential for enzymes, structure, transport, and regulation.

Key points

- Amino acids are absorbed or synthesized.
- Proteins undergo synthesis on ribosomes.
- Nitrogen assimilation is essential for amino acid formation.
- Nitrate reduction is an important step in plant nitrogen metabolism.

25. Nitrogen metabolism

Nitrogen is vital for amino acids, nucleotides, chlorophyll, and coenzymes.

Steps

- Nitrate uptake.
- Reduction to nitrite.
- Reduction to ammonium.
- Assimilation into amino acids.

Nitrogen fixation

Some microorganisms convert atmospheric nitrogen into usable forms, which is crucial for plant growth and soil fertility.

26. Secondary metabolism

Secondary metabolites are not directly essential for basic survival, but they help in defense, attraction, and ecological interaction.

Main groups

- Alkaloids.
- Terpenoids.
- Phenolics.
- Flavonoids.
- Tannins.
- Essential oils.

These compounds are very important in medicines, perfumes, dyes, and plant defense.

27. Stress physiology

Plants face environmental stresses such as drought, salinity, heat, cold, flooding, and pathogen attack.

Responses

- Osmotic adjustment.
- Closure of stomata.
- Production of stress proteins.
- Antioxidant defense.
- Hormonal changes.

Absciscic acid plays a major role in drought response.

28. Practical significance

Plant physiology and metabolism are important in:

- Crop improvement.
- Irrigation management.
- Fertilizer use.
- Greenhouse studies.
- Biotechnology.
- Plant breeding.
- Stress tolerance research.

29. Summary

Plant physiology explains how plants function, while metabolism explains the chemical basis of those functions. Together they describe water relations, nutrition, photosynthesis, respiration, growth, hormones, and stress responses, making this chapter central to advanced botany.